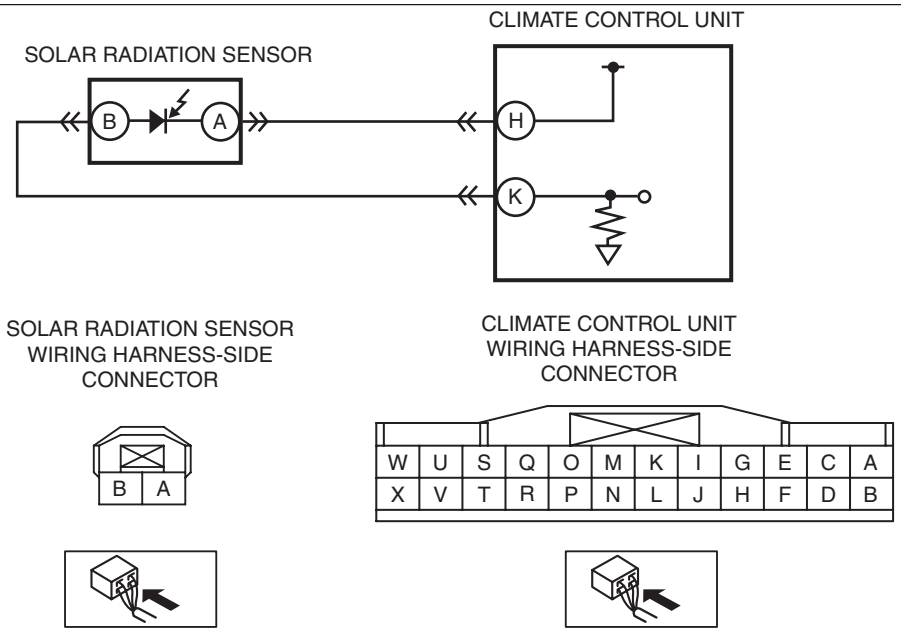
DTC B10BE:12, B10BE:13 [FULL-AUTO AIR CONDITIONER]

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DESCRIPTION	• B10BE:12 — Solar radiation sensor circuit short to power supply • B10BE:13 — Solar radiation sensor circuit open
DETECTION CONDITION	• Malfunction in wiring harness between solar radiation sensor and climate control unit
FAIL-SAFE FUNCTION	**Malfunction determined when IG SW at ON** • The solar radiation sensor input value is fixed at the value directly before the malfunction only when the valve is **1,450 W/m^2 or more**. **Malfunction already exists when IG SW turned to ON** • Solar radiation sensor value is fixed at **0 W/m^2** .
POSSIBLE CAUSE	• Solar radiation sensor connector malfunction • Solar radiation sensor malfunction • Open circuit in wiring harness between climate control unit and solar radiation sensor • Short to power supply in wiring harness between climate control unit and solar radiation sensor • Climate control unit connector malfunction • Climate control unit malfunction
SYSTEM WIRING DIAGRAM	

Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT SOLAR RADIATION SENSOR CONNECTOR** • Switch the ignition to off. • Disconnect the negative battery cable. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the solar radiation sensor connector. • Inspect the connector and terminals (corrosion, damage, pin disconnection). • Are the connector and terminals normal?	Yes	Go to the next step.
		No	Repair/replace the connector or terminal. After repair procedure, go to the next step.
2	**INSPECT SOLAR RADIATION SENSOR** • Inspect the solar radiation sensor. (See SOLAR RADIATION SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is it normal?	Yes	Go to the next step.
		No	Replace the solar radiation sensor. (See SOLAR RADIATION SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) Go to the next step.

STEP	INSPECTION		ACTION
3	**INSPECT SOLAR RADIATION SENSOR CIRCUIT FOR OPEN CIRCUIT** • Disconnect the climate control unit connector and the solar radiation sensor connector. • Is there continuity between the following terminals of the climate control unit and the solar radiation sensor? — H—A — K—B	Yes	Go to the next step.
		No	Repair the wiring harness. Go to the next step.
4	**INSPECT SOLAR RADIATION SENSOR CIRCUIT FOR SHORT TO POWER SUPPLY** • Connect the negative battery cable. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition to ON. • Measure the voltage of solar radiation sensor connector terminals A and B of solar radiation sensor harness side connector. • Is the voltage measured?	Yes	Repair the wiring harness. Go to the next step.
		No	Go to the next step.
5	**VERIFY CLIMATE CONTROL UNIT CONNECTOR CONDITION** • Switch the ignition to off. • Disconnect the negative battery cable. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect the connector and terminals (corrosion, damage, pin disconnection). • Are the connector and terminals normal?	Yes	Go to the next step.
		No	Repair/replace the malfunctioning vehicle wiring harness, connector, or terminal. After repair procedure, go to the next step.
6	**VERIFY THAT SAME DTC IS NOT OUTPUT AGAIN** • Reconnect the disconnected connectors. • Connect the negative battery cable. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition to ON. • Clear the DTC for the climate control unit using the M-MDS. (See CLEARING DTC [FULL-AUTO AIR CONDITIONER].) • Perform the DTC inspection for the climate control unit using the M-MDS. (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) • Is the same DTCs present?	Yes	Repeat the inspection from Step 1. • If the malfunction does not recur, go to the next step. • If the malfunction recurs, replace the climate control unit. (See CLIMATE CONTROL UNIT REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) Go to the next step.
		No	Go to the next step.
7	**VERIFY THAT NO OTHER DTCs ARE PRESENT** • Verify other DTCs displayed. • Are any other DTCs output?	Yes	Perform the corresponding DTC inspection.
		No	DTC troubleshooting completed.